

Question ID: b78cd5df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

An event planner is planning a party. It costs the event planner a onetime fee of **\$35** to rent the venue and **\$10.25** per attendee. The event planner has a budget of **\$300**. What is the greatest number of attendees possible without exceeding the budget?

Correct Answer: 25

Rationale

The correct answer is **25**. The total cost of the party is found by adding the onetime fee of the venue to the cost per attendee times the number of attendees. Let x be the number of attendees. The expression $35 + 10.25x$ thus represents the total cost of the party. It's given that the budget is **\$300**, so this situation can be represented by the inequality $35 + 10.25x \leq 300$. Subtracting **35** from both sides of this inequality gives $10.25x \leq 265$. Dividing both sides of this inequality by **10.25** results in approximately $x \leq 25.854$. Since the question is stated in terms of attendees, rounding **25.854** down to the greatest whole number gives the greatest number of attendees possible, which is **25**.